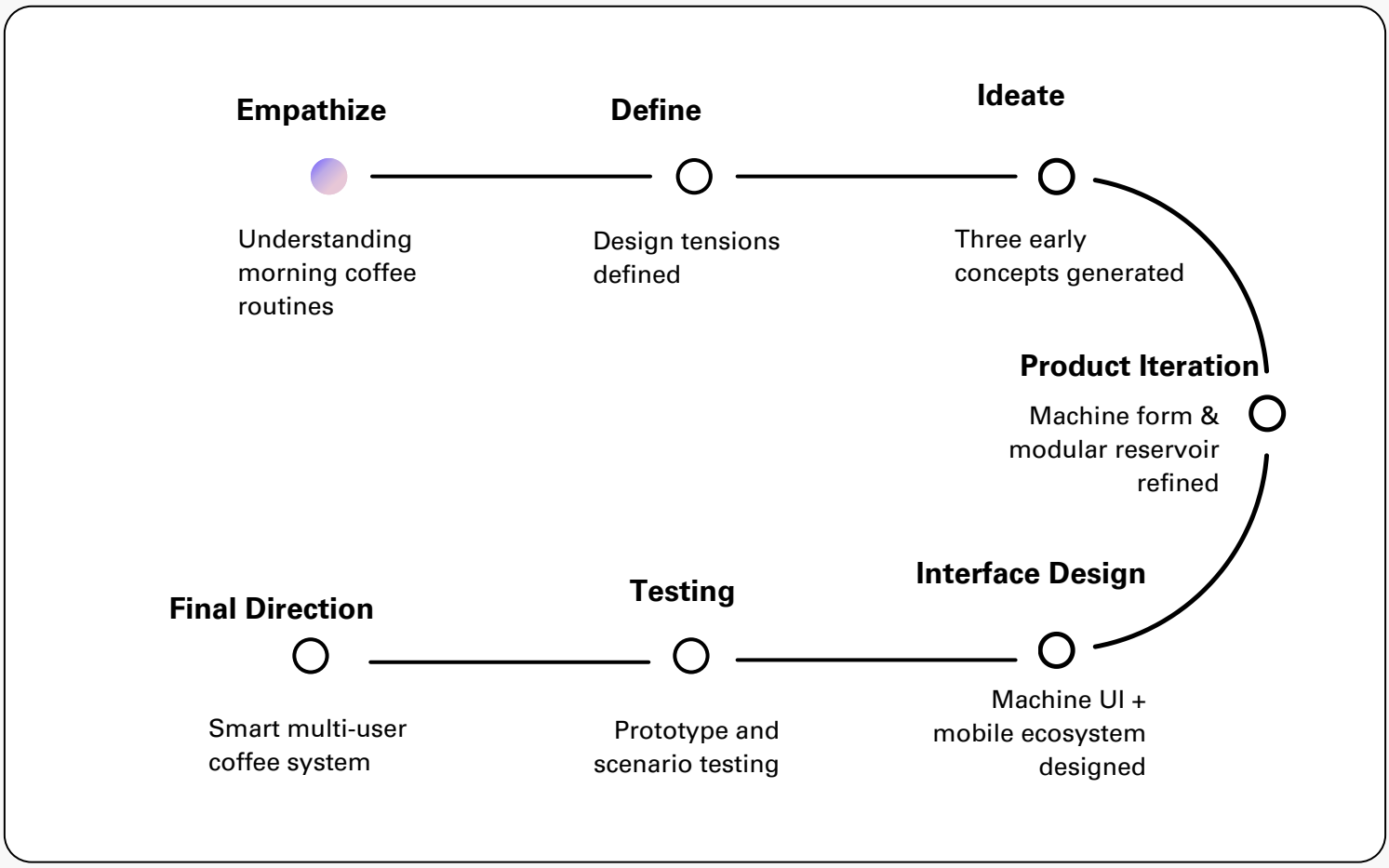
Selin Dengiz

Milestone Planning Roadmap



Empathize (User Research)

At the beginning of the project I explored how people prepare coffee in their daily routines, especially during **busy mornings.**

What I did

- Observed common problems with existing coffee machines
- Identified pain points in morning coffee preparation
- Analyzed how multiple users interact with a shared coffee machine

Key Insights

- Morning users want very fast interaction
- Most coffee machines require multiple input steps
- Brewing processes are not visible to the user
- In shared households, multiple users often want coffee at the same time

Define (Problem Framing)

Based on the research findings, I defined the main design problem.

Design Problem

How can a coffee machine reduce user interaction while making the brewing process more transparent and suitable for multi-user households?

Design Tensions

Quick coffee preparation
VS
Complex machine interaction
Process visibility
VS
Hidden machine operations

Ideate (Concept Generation)

During the ideation phase, multiple concept directions were explored.

Concept Directions

- Visible brewing machine
- Smart scheduling coffee machine
- Multi-user smart coffee system

Prototype (Concept Development)

Several product and interface prototypes were developed.

Product Iterations

Modular water reservoir concept
Integrated thermos output option
Visible coffee bean container
Smart indicator lights

Interface Prototypes

Machine interface screens:
Drink selection
Brew setup
Brewing process visibility
Multi-user queue
Warning notifications

Mobile application features:

Remote brewing
Scheduling
Household profiles
Saved coffee preferences

Test (Concept Evaluation)

The concept was tested through scenario simulations and usability reflection.

What Worked

One-time input interaction simplified the brewing process
Multi-user queue system supported shared households
Brewing stage visualization increased user trust

What Failed

Some users initially did not understand the removable reservoir

What Surprised Me

Users strongly appreciated seeing brewing stages such as grinding, heating, and pouring.

Keywords

- Busy morning routine
- Quick interaction
- Process visibility
- Multi-user household
- Smart brewing system
- Queue management
- One-time input
- Process transparency
- Smart kitchen ecosystem
- Visibility

Selected Impactful Prompts

Prompt 1
"Create a smart coffee machine concept for busy morning users with visible brewing and simplified interaction."

Prompt 2
"Generate a modular coffee machine with a removable water reservoir and visible coffee module."

Prompt 3
"Design a mobile app interface that allows multiple household users to send coffee commands."

Prompt 4
"Create a brewing process interface that visually shows grinding, heating, and brewing stages."

Prompt 5
"Create a 'Save Selected Items' feature that enables users to store their personal coffee preferences as profiles, allowing them to prepare their usual drink instantly each morning with one click."

Role	Description
Clarifier	Helped structure research insights and design tensions
Re-framer	Suggested alternative ways to interpret the design problem
Expander	Generated additional product features and interface ideas
Critique	Helped evaluate concept directions and refine decisions

AI Response Summaries

Clarifier
AI helped summarize research insights and identify the core tension between speed of interaction and machine complexity.

Re-framer
AI suggested reframing the machine from a simple appliance into a smart multi-user brewing system.

Expander
AI proposed additional features such as:

- brewing queue management
- smart notifications
- user profiles
- remote scheduling

Critique
AI helped compare early concepts and identify the one that best addressed the design tensions.

Reflections

Working with AI changed the way I explored design ideas. Instead of developing a single concept slowly, I was able to quickly generate and evaluate multiple alternatives. This accelerated the ideation stage and helped visualize complex systems such as the machine–app ecosystem.
AI also helped structure my design reasoning by turning vague ideas into clearer design decisions and interactions.

Persona

Name: Baki Yılmaz
Age: 29
Occupation: Product Designer
Location: Istanbul
Living Situation: Shares an apartment with two roommates

Background

Baki works full-time as a product designer and has a very busy morning routine. He usually wakes up shortly before leaving for work and prefers a quick and efficient way to prepare his coffee. Since he shares the kitchen with roommates, multiple people often want coffee at the same time, which sometimes creates delays.

- Goals
- Prepare coffee as quickly as possible in the morning
- Avoid complex machine settings
- Get consistent coffee with minimal interaction
- Be able to schedule or send commands remotely

Frustrations

- Coffee machines that require too many steps
- Waiting while someone else uses the machine
- Not knowing what the machine is doing during brewing
- Running out of water or beans without warning

Needs

- Fast one-time input interaction
- Visible brewing process
- Multi-user support
- Smart notifications
- Saved personal coffee preferences

Morning Scenario

07:10 – Baki wakes up and opens the coffee machine app.
07:11 – He selects his saved coffee profile.
07:12 – The machine adds his coffee to the brew queue.
07:14 – The machine prepares his Americano.
07:16 – Coffee is ready in the thermos and Baki leaves for work.

Design Decisions

Decision 1
Focus on one-time input interaction to reduce repeated commands during morning routines.
Decision 2
Introduce a multi-user queue system to support shared households.
Decision 3
Make the brewing process visually transparent through the interface.
Decision 4
Optimize the machine for countertop space efficiency with a modular transparent water reservoir and transparent coffee module.

What Was Accepted, Rejected, and Transformed

Accepted

✓ One-time brew setup interaction
✓ Multi-user queue system
✓ Process visibility during brewing
✓ Modular water reservoir and transparent coffee module design

Rejected

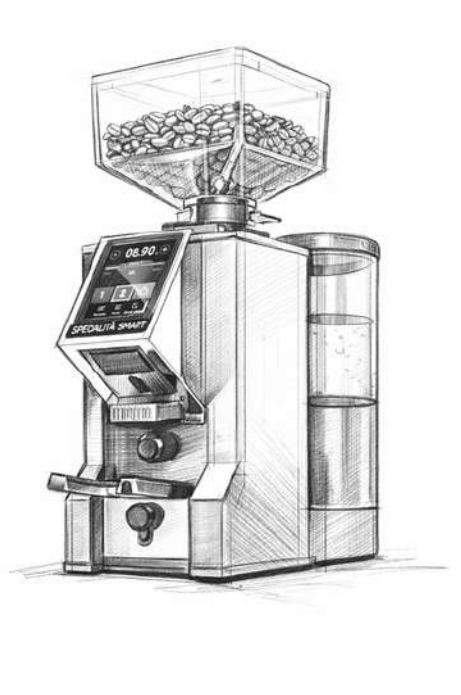
✕ Complex multi-screen input interaction
✕ Hidden brewing process with minimal feedback
✕ Fixed non-removable water tank

inal Design Direction

The final concept is a smart multi-user coffee machine designed for busy households. The system simplifies coffee preparation by enabling users to send a single brewing command, while the machine automatically manages preparation order, brewing stages, and notifications. The design aims to minimize user input and allow the coffee preparation process to be completed as quickly and efficiently as possible, supporting fast morning routines and reducing interaction time.

The design focuses on:

- reducing interaction time
- increasing process transparency
- supporting multiple users



Water 85%
Beans 62%
07:15
Choose Drink
Select your beverage
Profiles
Select user
Schedule
Set brewing time

Water
Water 85%
Beans 62%
07:15
Black Coffee
Espresso
Strong coffee shot
Americano
Espresso with hot water
Americano
Longer espresso shot
Café Cortado
Espresso with a dash of milk
Milk-Based
Cappuccino
Espresso, steamed milk milk & foam
Latte Macchiato
Larger size latte machiato

Water
Water 85%
Beans 62%
07:15
Brew Setup
Americano
Mild
Medium
Strong
Servings
1
2
3
4
4
Timing
Brew now
Thermos
Save as favorite
OFF
Save as favorite
OFF
CONFIRM BREW

Water 85%
Beans 62%
07:15
Household Profiles
Baki
Americano · Medium · 1 serving
07:15
Ayşe
Cappuccino · Mild · 1 serving
Ali
Espresso · Strong · 1 serving
CONFIRM BREW

09:12
07:13
BREW QUEUE
Ayse - Latte
Baki - Americano
Estimated start
2 min
VIEW DETAILS

09:12
07:13
Brewing Status
Grinding
Heating
Brewing
Pouring
Ready
68%
Estimated finish 01:20
Baki - Americano
Container - Thermos
VIEW DETAILS

A smart thermos designed to work with the coffee machine detects when the coffee has reached a drinkable temperature. When the coffee is ready to drink, a circular light around the thermos turns on to notify the user.